1 Aura Theorem

In our proof of the Intermediate Value Theorem, we will use the following fact about continuous functions.

Theorem 1.1 (Aura Theorem)**.** *Let f be continuous at a.*

(a) If $f(a) > 0$, then $f(x) > 0$ for all x in some open interval containing a.

(b) If $f(a) < 0$, then $f(x) < 0$ for all x in some open interval containing a.

Proof. We prove part (a) and leave part (b) as an exericse. Assume $f(a) > 0$. Then corresponding to $\frac{f(a)}{2} > 0$ there exists a corresponding $\delta > 0$ such that $|x-a| < \delta$ implies

$$|f(x)-(a)| < \frac{f(a)}{2} \iff -\frac{f(a)}{2} < f(x)-f(a) < \frac{f(a)}{2} \iff 0 < \frac{f(a)}{2} < f(x) < \frac{3f(a)}{2},$$

hence $f(x) > 0$ for all $x \in (a-\delta, x+\delta)$. □

2 Preliminary results

Recall the following preliminary results from Unit 1.

Capture Theorem. Let A be a subset of $\mathbb{R}$. Any open interval containing $\sup A$ contains an element of A. Similarly, any open interval containing $\inf A$ contains an element of A.

Aura Theorem. Let f be continuous at a.

(a) If $f(a) > 0$, then $f(x) > 0$ for all x in some open interval containing a.

(b) If $f(a) < 0$, then $f(x) < 0$ for all x in some open interval containing a.

Nested Intervals Theorem. For each $n \in \mathbb{N}$, assume we are given a closed interval $I_n = [a_n, b_n] = \{x \in \mathbb{R} : a_n \leq x \leq b_n\}$ such that

(i) $a_n \leq a_{n+1}$ and $b_{n+1} \leq b_n$ for all $n \in \mathbb{N}$, so that $I_1 \supseteq I_2 \supseteq I_3 \supseteq I_4 \supseteq \cdots$ (see the picture below).

Then $\boxed{\bigcap_{n=1}^{\infty} I_n \neq \emptyset}$.

(ii) If, in addition, $\inf\{b_n - a_n\} = 0$,

then

$$\boxed{\bigcap_{n=1}^{\infty} I_n = \{x\}, \text{ where } x = \sup\{a_n\} = \inf\{b_n\}}.$$

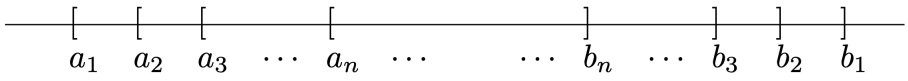


Figure 1: Intervals satisfying the hypotheses of the Nested Intervals Theorem.

Bisection Procedure.

- Begin with a closed interval $I_1 = [a_1, b_1]$.
- Bisect I_1 to obtain two closed subintervals $[a_1, \frac{a_1+b_1}{2}]$ and $[\frac{a_1+b_1}{2}, b_1]$.
- Choose one of the two subintervals above, and call it $I_2 = [a_2, b_2]$.
- Repeating this process indefinitely results in a nested sequence of intervals

  $$I_1 \supseteq I_2 \supseteq I_3 \supseteq I_4 \supseteq \cdots$$

  with $\inf\{b_n - a_n\} = 0$.
- The Nested Intervals Theorem then implies $\bigcap_{n=1}^{\infty} I_n = \{x\}$, where $x = \sup\{a_n\} = \inf\{b_n\}$.

3 Proof of the Intermediate Value Theorem

We first prove a special case of the Intermediate Value Theorem using the Bisection Procedure. The general case then follows easily.

Theorem 3.1 (Bolzano's Theorem)**.** *Let f be a continuous function defined on $[a,b]$. If 0 is between $f(a)$ and $f(b)$, then there exists $x \in [a,b]$ such that $f(x) = 0$.*

Proof.

1. If $f(a) = 0$ or $f(b) = 0$ we are done. Suppose now that $f(a) < 0 < f(b)$. Let $I_1 = [a_1, b_1] = [a,b]$. Consider the midpoint of I_1, $\frac{a_1+b_1}{2}$. If $f(\frac{a_1+b_1}{2}) = 0$, we are done. Suppose now that $f(\frac{a_1+b_1}{2}) \neq 0$. Explain why this implies f changes sign at the endpoints of one of the two subintervals intervals $[a_1, \frac{a_1+b_1}{2}]$ or $[\frac{a_1+b_1}{2}, b_1]$.

2. Choose the subinterval above on which f changes sign and call it $I_2 = [a_2, b_2]$. Explain how to choose the next subinterval $I_3 = [a_3, b_3]$.

3. Continue as above to obtain a collection of subintervals $\{I_n\}$ satisfying the hypotheses of the Nested Intervals Theorem. The Nested Intervals Theorem then implies $\bigcup_{n=1}^{\infty} I_n = \{x\}$, where $x = \sup\{a_n\} = \inf\{b_n\}$. Suppose $f(x) > 0$. What does the Aura Theorem tell you?

4. Now what does the Capture Theorem tell you?

5. Explain why $f(x) > 0$ leads to a contradiction.

6. Show that $f(x) < 0$ also leads to a contradiction.

7. What do you conclude?

8. Show that the conclusion of Bolzano's Theorem also holds if $f(b) < 0 < f(a)$. (*Hint*: Consider the function $g(x) = -f(x)$.)

Theorem 3.2 (Intermediate Value Theorem)**.** *Let f be a continuous function defined on $[a,b]$. Then for any c between $f(a)$ and $f(b)$ there exists $x \in [a,b]$ such that $f(x) = c$.*

9. Use the result of Bolzano's Theorem to prove the Intermediate Value Theorem.
(*Hint:* If $f(a) < c < f(b)$, consider the function $g(x) = f(x) - c$.)